# Messy master data hides money. Tesseract fixes it at the front door.

## 1 PROBLEM

### Amy needed one item code. It took 18 days.

Our breakdown of App C (illustrative): before → after

| | |
|---|---|
| Waiting for MDM to notice the email | 2 → 0 d |
| Bounced for the missing QA/RA field, plus travel delay | 5 → 0 d |
| **Three approvers one at a time (~3 days each)** | 9 → 3 d |
| MDM queue and keying into SAP by hand | 2 → 1 d |

**18 → ~4 calendar days.** ≈43% of the saving is a process change: approve in parallel, no new software.

**Incomplete and duplicate records**
Requests arrive by email; 40% bounce, 6% of new customers already exist (A13).

**Invisible approvals, no audit trail**
Approvers sign one at a time; nobody can see where a request is stuck.

**Regional numbers no one trusts**
The same hospital sits under 3 codes, so growth and decline are hidden.

Fact: SAP B1's built-in approvals are document-centric

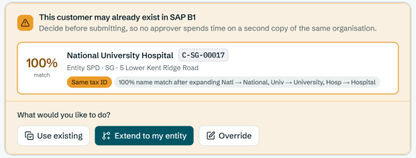
## 2 SOLUTION

### One front door on Microsoft 365, landing in SAP Business One

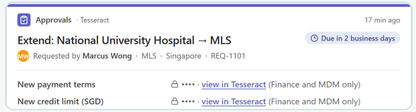
| | | |
|---|---|---|
| Requestor | Power Apps form | Existing licences |
| Checks | Duplicate + field check | Standard connectors |
| Approvers | Teams approvals | Standard connector |
| MDM | SharePoint List | Existing licences |
| SAP B1 | DTW import file | S$0 · ships with B1 |
| Reporting | Regional Pulse | Power BI Pro if not owned |

MDM team configures · Lean team builds · Outside help

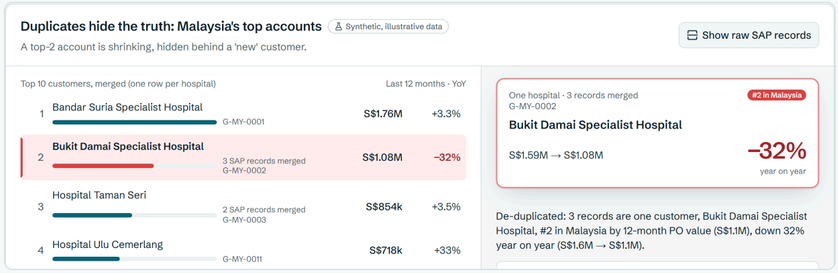
**Phase 2 (not built):** automatic SAP posting, ≤USD 1,980/yr




Duplicate caught before approval




Approve in Teams




Regional Pulse: duplicates merged, a hidden decline appears

## 3 WHY THIS, NOT...

### The only option that fixes both the workflow and the numbers

| Option | Cost | Stops incomplete | Stops duplicates | Fast approvals | MDM-owned, no IT | Regional analytics |
|---|---|---|---|---|---|---|
| Email template + tracker | ~S$0 | ~ | ✗ | ✗ | ✓ | ✗ |
| SAP B1 approval procedures | S$0 | ✗ | ✗ | — | — | ✗ |
| Teams Approvals only | ~S$0 | ~ | ✗ | ✓ | ✓ | ✗ |
| Power Platform workflow only | ~S$30k | ✓ | ~ | ✓ | ✓ | ✗ |
| Buy an MDM suite / B1 add-on | High | ✓ | ✓ | ✓ | ~ | ~ |
| **Tesseract** | ~S$40k | ✓ | ✓ | ✓ | ✓ | ✓ |

✓ yes · ~ partly · ✗ no · — not applicable

**SAP B1 approvals** are document-centric: no structured intake, no duplicate check, no parallel SLA for master-data requests.

**Teams Approvals alone** is a good zero-build first step, but has no duplicate check and no SAP-ready output.

# Impact, cost and how we get it adopted

## Median 14 days → ≤ 4 business days

Setup time for a new record (A26). Amy's took 18; ≈43% of the saving is a process change.

## ≈1,750 hours a year

1,751 h (≈0.94 FTE) of analyst, requestor and MDM time handed back. Capacity, not headcount.

## Payback ≈ 6 months

5.9 months on a S$39.9k phase-1 one-off, hard hours only, at 80 requests a month.

## No new licences

for the workflow in phase 1: standard Microsoft 365 connectors, Teams, SharePoint and SAP B1's own import tool.

### BUSINESS CASE (HARD HOURS × S$50/H)

| Requests / month | Hours / yr | Net / yr | Payback | 3-yr net |
|---|---|---|---|---|
| 40 | 1,115 | S$49.8k | 9.6 mo | S$109k |
| **80 (base)** | **1,751** | **S$81.5k** | **5.9 mo** | **S$205k** |
| 120 | 2,386 | S$113.3k | 4.2 mo | S$300k |

**Phase 1: S$39.9k one-off + S$6.0k/yr run**
Lean team 2 × 6 weeks, a 15-day contractor, MDM config, 30-min onboarding. Cash ≈ the contractor; the rest is internal time.

**Phase 2, only after the pilot: S$22.5k + ≤S$2.6k/yr licences**
Automatic SAP posting saves only the import step: ~19-year payback on hours, so it is a speed decision.

**Upside, not counted**
EDGE grant ≈S$16k if Lives qualifies as an SME; approver time. Opportunity figures are never added to payback.

### PILOT, KPIS AND ADOPTION

**6-week pilot · SG + MY · Customer + Item domains · SPD + QTS**

| | |
|---|---|
| **≥90%** | of requests through Tesseract by week 6 |
| **≤4** | business days median setup (baseline 14 days) |
| **<15%** | bounce rate (baseline 40%) |
| **Count** | duplicates stopped at intake, weekly |
| **Falling** | S$ of orders waiting on a setup |

**Week 6 gate:** median >6 business days or adoption <70% → fix or stop. KPIs met → TH, VN, MM, then Vendor and Cost centre.

**Adoption:** one front door (the MDM mailbox auto-replies with the form link) · one-click approvals in Teams · visible status ends chasing · a champion per entity · weekly SLA scoreboard.

### REGIONAL INSIGHT (ILLUSTRATIVE DATA)

Merging 25 duplicate records (of 388) shows Bukit Damai Specialist Hospital, Malaysia's #2 account, is down 32%.

S$1.34M of orders waited on an MDM setup in the last 12 months, 15.4 days on average.

On our illustrative data, today's method is off by 11 units per 100 ordered; ours by 9 (6 unseen months, rising dengue season).

**Not in v1, by design:** automatic SAP posting, merging existing SAP duplicates, deep-learning forecasts.

**Roadmap:** M365 pilot build → vendor bank-detail changes with dual Finance control → delegation and time-zone SLAs → Power BI Pulse on real B1 extracts.

**Labelled assumptions** (full register at /assumptions): **A1** 5 countries: SG, MY, TH, VN, MM · **A5/A27** Oct 2025–Sep 2026 vs the year before; 36 months of synthetic history (B1 holds ≥24) · **A8** growth = 12-month YoY PO value on de-duplicated customers · **A9** independent approvals run in parallel (policy change) · **A12–A13** ~80 requests/month; 40% bounce, 6% duplicate customers · **A17** all data synthetic, names fictional · **A19–A26** 1.25 h saved per request, 4 h per duplicate, 40 h/month consolidation, S$50/h, build and partner costs (low confidence), 10% leak, USD 1 = S$1.33, 14-day baseline.

**CTFudders · SMU**
**DSA 2026**

Live prototype
**onerecord-lives-dsa26.vercel.app**

Code (GitHub)
**github.com/ethantyh03/onerecord**

Demo video (YouTube)
**youtu.be/5tieuSmd2Aw**